

Unit 3.3 - Integrals

Contents

1 Introduction to Integration

Integration is a fundamental concept in calculus that can be understood in two complementary ways:

- As the **antiderivative**, the reverse process of differentiation.
- As the calculation of the **area under a curve**.

These two perspectives are deeply connected by the *Fundamental Theorem of Calculus*, which shows that they are, in essence, two sides of the same coin.

1.1 Integration as an Antiderivative

One way to think about integration is as the reverse of differentiation. If $f(x)$ represents the rate of change of some function $F(x)$, then the integral of $f(x)$

$$F(x) = \int f(x) dx$$

gives us the original function $F(x)$, up to a constant (more on this in the footnote below). The above equation means essentially the same thing as the equation

$$F'(x) = f(x)$$

i.e., that the two functions are related by a derivative/antiderivative relationship.

As one might expect, the *antiderivative* is the opposite of a derivative. Instead of finding the rate of change of a function f , the antiderivative finds a function F whose rate of change is f , the original function. In general, the antiderivative of a function is represented with a capital letter version of the function (thus, the choices of f and F).

This approach allows us to reconstruct a function from its rate of change, which is useful in many real-world scenarios. For example, we can calculate the distance some object travelled if provided a function $v(t)$ which represents its

velocity (the rate of change of distance or position) over time.¹ In the simplest case where the velocity is constant ($v(t) = c$), then we can simply multiply that velocity times the duration of travel Δt :

$$\Delta x = c\Delta t$$

In fact, this simply represents the area under the graph of the velocity function. This moves us nicely into the next interpretation of integration.

1.2 Integration as Area Under the Curve

Another interpretation of integration is geometric: it calculates the area under a curve. Given a function $f(x)$, the integral from a to b ,

$$\int_a^b f(x) dx,$$

represents the area enclosed by the curve $y = f(x)$, the x -axis, and the vertical lines $x = a$ and $x = b$. (NOTE: when an integral has limits a, b at the top and bottom of the integral sign, it is called a *definite integral*; but when it has not limits, it is called an *indefinite integral*.) Figure ?? depicts this definite integral graphically.

1.3 The Fundamental Theorem of Calculus

The **Fundamental Theorem of Calculus** provides a key connection between these two views of integration: antiderivatives and areas under curves. It states that if $f(x)$ is continuous on the interval $[a, b]$, then:

$$\int_a^b f(x) dx = F(b) - F(a).$$

i.e., that the total area under the curve from a to b can be found by evaluating the antiderivative $F(x)$ at the endpoints. Additionally, we can replace f with its derivative to get the following:

$$\int_a^b f'(x) dx = f(b) - f(a).$$

thus, showing how the integral relates to the derivative, as well.

¹However (and this relates back to the above statement: up to a constant), doing this only let's us calculate the total distance travelled, not the object's absolute position. If we wished to know the object's absolute final position, we would need to know where it started its journey and then ADD the total distance travelled. This initial position is what is referred to by the *constant* in that statement. Unfortunately, performing an integral/antiderivative on this velocity function doesn't grant us knowledge of this addition constant that is the object's starting point. It is an extra piece of information which must be known via other means. When we think about it, this notion of an unknown constant when performing integration makes intuitive sense when thought in light of the derivative. Suppose we took the derivative of a function $f(x)$. The would eliminate and constant terms in the equation. Thus, when we tried to take an integral/antiderivate to get the original function back, we wouldn't know what constant to use since literally any choice of constant would be eliminated by the derivative process.

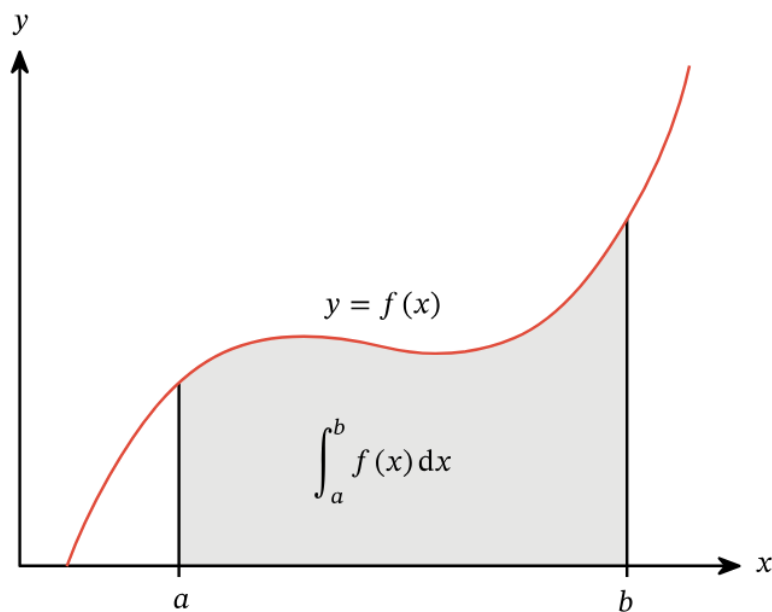


Figure 1: Visualization of the Definite Integral as the Area Under a Curve. (FROM: Nagwa)

1.4 The Riemann Sum

When it comes to actually computing definite integrals, we can utilize something called a *Riemann sum*. This is a technique which divides the area under the function's curve into many smaller areas which are then added together (see Figure ??).

The Riemann sum involves splitting the interval $[a, b]$ into n subintervals of equal width:

$$\Delta x = \frac{b - a}{n}.$$

For each subinterval, we approximate the area under the curve by using the value of the function at some point x_i^* within that subinterval, often chosen as the left endpoint, right endpoint, or the midpoint. The total sum of these small areas provides an approximation for the integral:

$$S_n = \sum_{i=1}^n f(x_i^*) \Delta x,$$

As the number of subintervals n increases, the width of each subinterval Δx becomes smaller, and the Riemann sum becomes a more accurate approximation of the area under the curve. In the limit as n approaches infinity, the Riemann

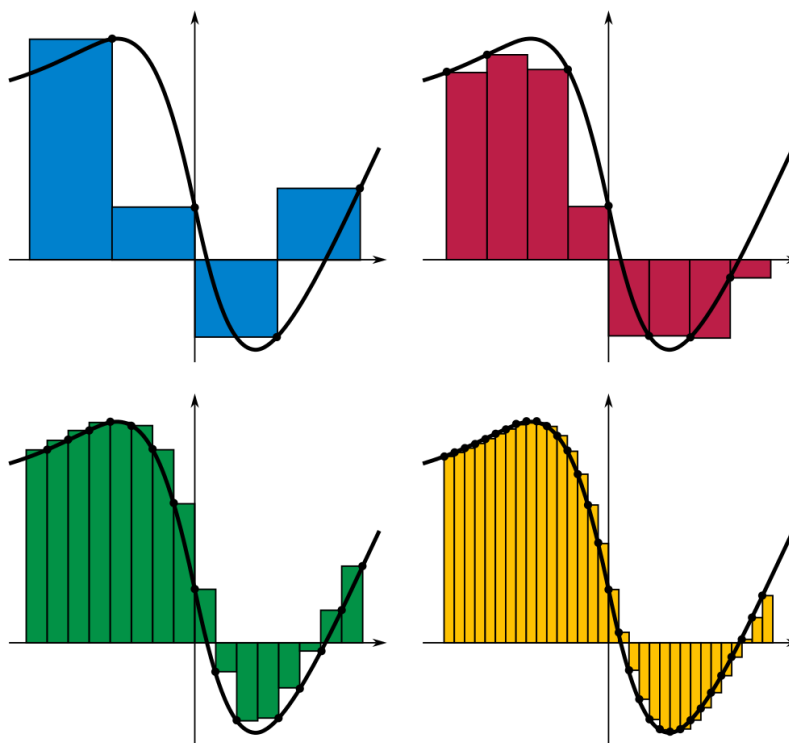


Figure 2: Visualization of the Riemann sum with different n . (FROM: Wikipedia)

sum converges to the exact value of the definite integral:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

2 Numerical Integration

We will now move on to taking definite integrals numerically. In fact, the definition of the definite integral in terms of the Riemann sum is easily adapted to numerical applications.

2.1 Numerically Integrating a Function

When numerically integrating a continuous function $f(x)$, we approximate the area under the curve by truncating the infinite Riemann sum to a finite n .

Regarding the choice of x_i^* , there are numerous methods. Here are four common techniques (see Figure ?? for graphical depictions of these rules):

- **Left-Hand Rule:** In this method, we approximate the area under the curve by using the function value at the left endpoint x_i of each subinterval:

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} f(x_i) \Delta x$$

where $\Delta x = \frac{b-a}{n}$ is the width of each subinterval.

- **Right-Hand Rule:** Here, we approximate the area under the curve by using the function value at the right endpoint x_{i+1} of each subinterval:

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} f(x_{i+1}) \Delta x$$

- **Midpoint Rule:** The function is evaluated at the midpoint $\frac{x_i + x_{i+1}}{2}$ of each subinterval, providing a more accurate approximation compared to the left- and right-hand rules:

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} f\left(\frac{x_i + x_{i+1}}{2}\right) \Delta x$$

- **Trapezoidal Rule:** This method uses trapezoids instead of rectangles to approximate the area under the curve, averaging the left- and right-hand function values:

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} \left(\frac{f(x_i) + f(x_{i+1})}{2} \right) \Delta x$$

As we increase the number of subintervals n , or equivalently decrease the subintervals' width Δx , the approximation becomes more accurate. However, there is a limit to this improvement due to floating-point precision. When Δx becomes too small, round-off errors can accumulate, leading to inaccuracies. As a result, there is an optimal number of subintervals where the balance between approximation accuracy and computational precision is best achieved. This optimal number and width is problem dependent.

2.2 Numerically Integrating a Fixed Dataset

In the case of numerical integration with a fixed dataset, we no longer have the flexibility to choose arbitrary subintervals. The points of the dataset define the intervals. For a dataset given by $\{(x_1, y_1), \dots, (x_n, y_n)\}$, we apply the same four rules as follows:

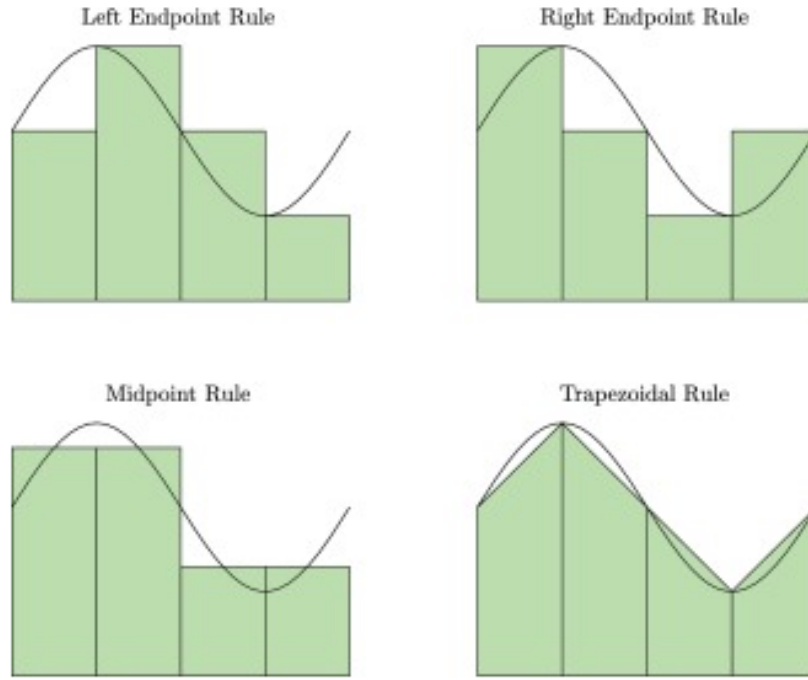


Figure 3: Visualization of the four numerical integration rules. (FROM: ScienceDirect)

- **Left-Hand Rule:** The approximation uses the data point at the left point of each interval (y_i):

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} y_i \cdot (x_{i+1} - x_i)$$

- **Right-Hand Rule:** This method uses the right point of each interval (y_{i+1}):

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} y_{i+1} \cdot (x_{i+1} - x_i)$$

- **Midpoint Rule:** For the midpoint rule, we need to interpolate the values between data points. If we assume *linear interpolation*, the midpoint rule becomes equivalent to the trapezoidal rule. More accurate approximations can be obtained using higher order polynomials like quadratics or cubics. If we let $I(x, y_i, y_{i+1})$ be a function which interpolates between the values y_i, y_{i+1} at some given point x , the midpoint rule becomes:

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} I\left(\frac{x_{i+1} + x_i}{2}, y_i, y_{i+1}\right) \cdot (x_{i+1} - x_i)$$

- **Trapezoidal Rule:** This rule is directly applicable to the dataset without interpolation, as it uses the values at the data points:

$$\int_a^b f(x) dx \approx \sum_{i=1}^{n-1} \left(\frac{y_{i+1} + y_i}{2} \right) \cdot (x_{i+1} - x_i)$$

Again, this is identical to the midpoint rule when linear interpolation is used.

As stated above, unlike the case of continuous functions, we cannot reduce the step size arbitrarily when dealing with datasets, as the interval widths $x_i - x_{i-1}$ are fixed. The accuracy of the integration depends on the spacing of the data points. Interpolation can be used to estimate values between points and increase accuracy to a degree, but it cannot replace the precision of having more data points.